

# TikZ Figure Template

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## About this template

This document draws a native TikZ/pgfplots figure inside a floating **figure** environment and references it with raw LaTeX `\ref{fig:parabola}`. No `pandoc-crossref` filter is required.

## TikZ via pgfplots

See Figure 1 for a parabola rendered natively by pgfplots.

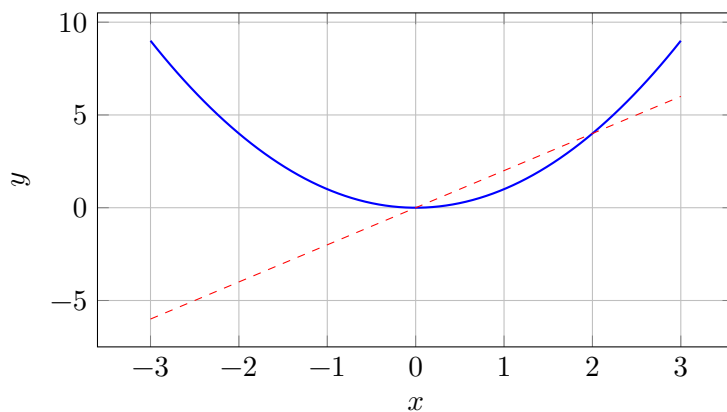


Figure 1: Parabola  $y = x^2$  (blue) with its derivative  $y = 2x$  (red).

## Plain TikZ

A minimal TikZ picture without pgfplots, for completeness:

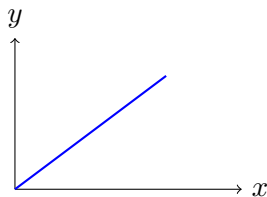


Figure 2: Basic TikZ axes with a vector.

See also Figure 2.

## Notes

- `{=latex}` raw blocks are preserved verbatim when targeting LaTeX/PDF; they are dropped for HTML output (that is intentional — see `strip-raw-latex-for-html.lua` if HTML+TikZ becomes a requirement).
- `\pgfplotsset{compat=1.18}` freezes pgfplots' internal behavior; bump the version if newer features are needed.
- TikZ compiles with `xelatex`. For very heavy TikZ jobs, `lualatex` is an acceptable fallback (CON-002).

## Compile

```
pandoc tikz-figure.md \
  --pdf-engine=xelatex \
  -o tikz-figure.pdf
```